



Vector Derivation

Department of Computer Engineering
Sharif University of Technology

Maryam Ramezani maryam.ramezani@sharif.edu

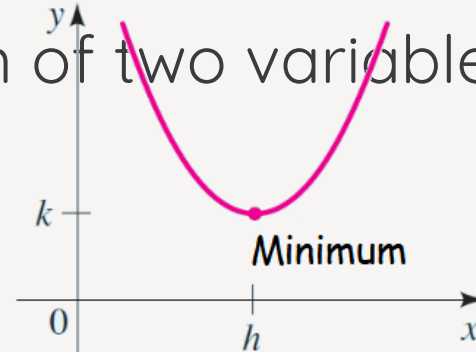
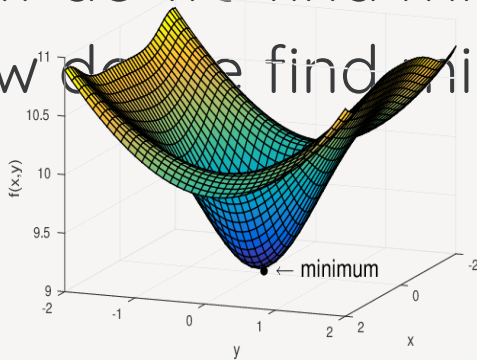
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Introduction



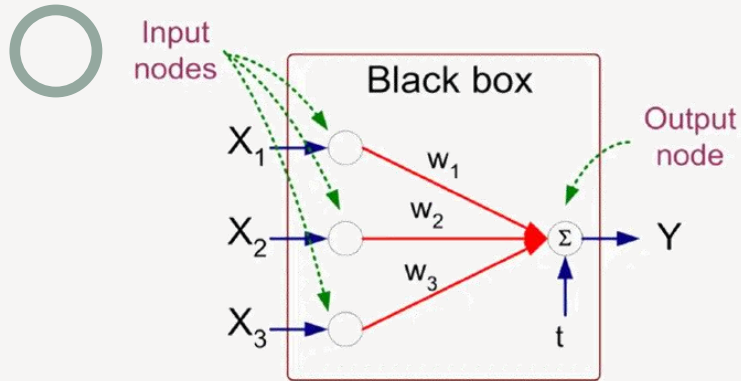
Motivation

- ❑ Machine Learning training requires one to evaluate how one vector changes with respect to another?
- How output changes with respect to parameters?
- How do we find minimum of a scalar function?
- How do we find minimum of two variables?

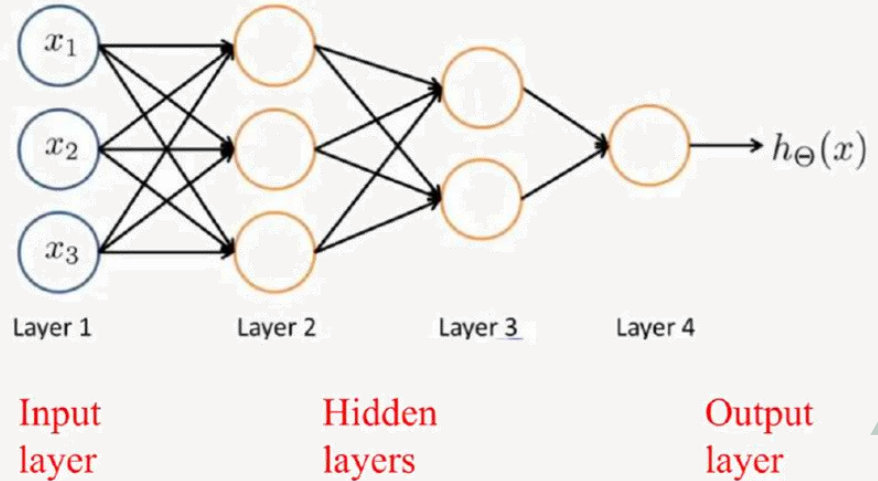


Neural Network

A single neuron

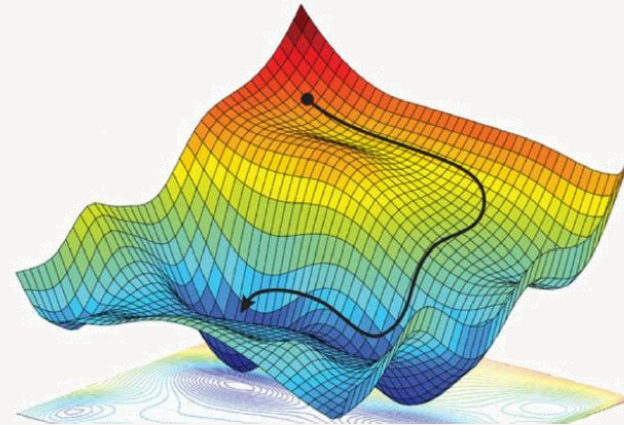
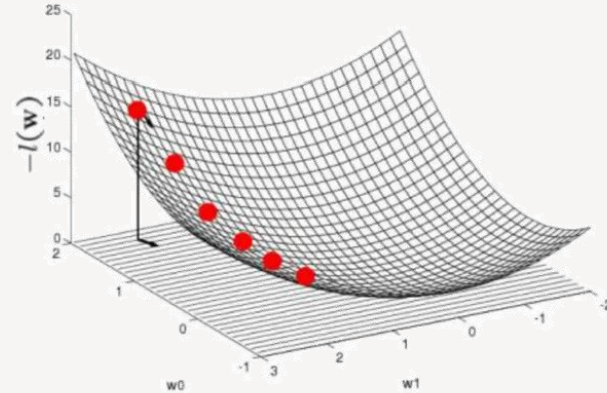


A neural network



ML Optimization

- Optimizing the weights of a neural network, or more generally the parameters of a machine learning model, can be an extremely complex task.
- Many tools have been developed for this purpose. The core of these tools relies on the use of "local information," such as derivatives (gradients) and similar methods.
- Here, the problem is to search for and find the optimal weights in a continuous space, which has an infinite number of potential candidates. Such a problem is also referred to as Continuous



Different Functions

- Scalar Function $f: \mathbb{R} \rightarrow \mathbb{R}$
- Scalar Field $f: \mathbb{R}^n \rightarrow \mathbb{R}$ or $f: \mathbb{R}^{n \times k} \rightarrow \mathbb{R}$ or $f: \mathbb{R} \rightarrow \mathbb{R}$
- Vector Field $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$ or $f: \mathbb{R}^{n \times k} \rightarrow \mathbb{R}^m$ or $f: \mathbb{R} \rightarrow \mathbb{R}^m$
- Matrix Field $f: \mathbb{R}^n \rightarrow \mathbb{R}^{n \times m}$ or $f: \mathbb{R}^{n \times k} \rightarrow \mathbb{R}^{p \times m}$ or $f: \mathbb{R} \rightarrow \mathbb{R}^{p \times m}$
- Tensor Field $f: \text{scalar, vector, matrix} \rightarrow \mathbb{R}^{n \times m \times k}$

In higher dimensions, if we take the derivative of a scalar field, it will result in a scalar field (Gradient). If we take the derivative again, it will result in a matrix-valued function (Hessian).

Overview

Types of matrix derivative

Types	Scalar	Vector	Matrix
Scalar	$\frac{\partial y}{\partial x}$	$\frac{\partial \mathbf{y}}{\partial x}$	$\frac{\partial \mathbf{Y}}{\partial x}$
Vector	$\frac{\partial y}{\partial \mathbf{x}}$	$\frac{\partial \mathbf{y}}{\partial \mathbf{x}}$	Tensor! (Optional part of this course)
Matrix	$\frac{\partial y}{\partial \mathbf{X}}$		

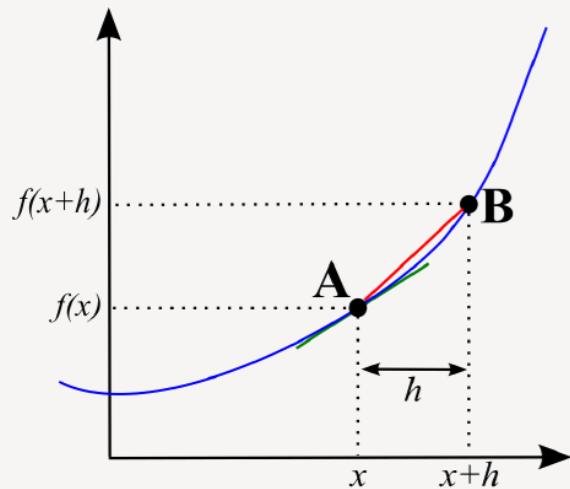
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Scalar function Derivation



Overview

$$f'(x) \text{ or } \frac{df}{dx}(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$



- ❑ A derivative, which itself is a function $f': \mathbb{R} \rightarrow \mathbb{R}$, stores local/instantaneous information about changes in the function.
- ❑ Note that the derivative may not be defined at certain points (or anywhere at all). Functions that are differentiable throughout their domain are referred to as **differentiable**.

Simple Rules

1. Constant Rule : $\frac{d}{dx}(c) = 0$
2. Constant Multiple Rule : $\frac{d}{dx}[cf(x)] = cf'(x)$
3. Power Rule : $\frac{d}{dx}(x^n) = nx^{n-1}$
4. Sum Rule : $\frac{d}{dx}[f(x) + g(x)] = f'(x) + g'(x)$
5. Difference Rule : $\frac{d}{dx}[f(x) - g(x)] = f'(x) - g'(x)$
6. Product Rule : $\frac{d}{dx}[f(x)g(x)] = f(x)g'(x) + g(x)f'(x)$
7. Quotient Rule : $\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{g(x)f'(x) - f(x)g'(x)}{[g(x)]^2}$
8. Chain Rule : $\frac{d}{dx}f[g(x)] = f'[g(x)]g'(x)$

Well-Behaved Functions in Differentiation

A function is considered well-behaved if it satisfies these criteria:

- Continuity: The function is continuous across its domain (no jumps or breaks).
- Differentiability: The function is differentiable at every point in its domain (no sharp corners).
- Smoothness: The derivative is also continuous, ensuring smooth transitions.



Well-Behaved Functions in Differentiation

Examples of Well-Behaved Functions

- Polynomials: $f(x) = x^2$, $f(x) = 3x^3 + 2x - 5$
- Trigonometric: $f(x) = \sin(x)$, $f(x) = \cos(x)$
- Exponential: $f(x) = e^x$
- Logarithmic (defined domain): $f(x) = \ln(x)$, $x > 0$

Non-Well-Behaved Functions

- Discontinuous: $f(x) = \frac{1}{x}$ at $x = 0$
- Sharp Points: $f(x) = |x|$ at $x = 0$
- Oscillatory: $f(x) = x \sin(1/x)$ at $x = 0$

Interpretation of First and Second Derivatives

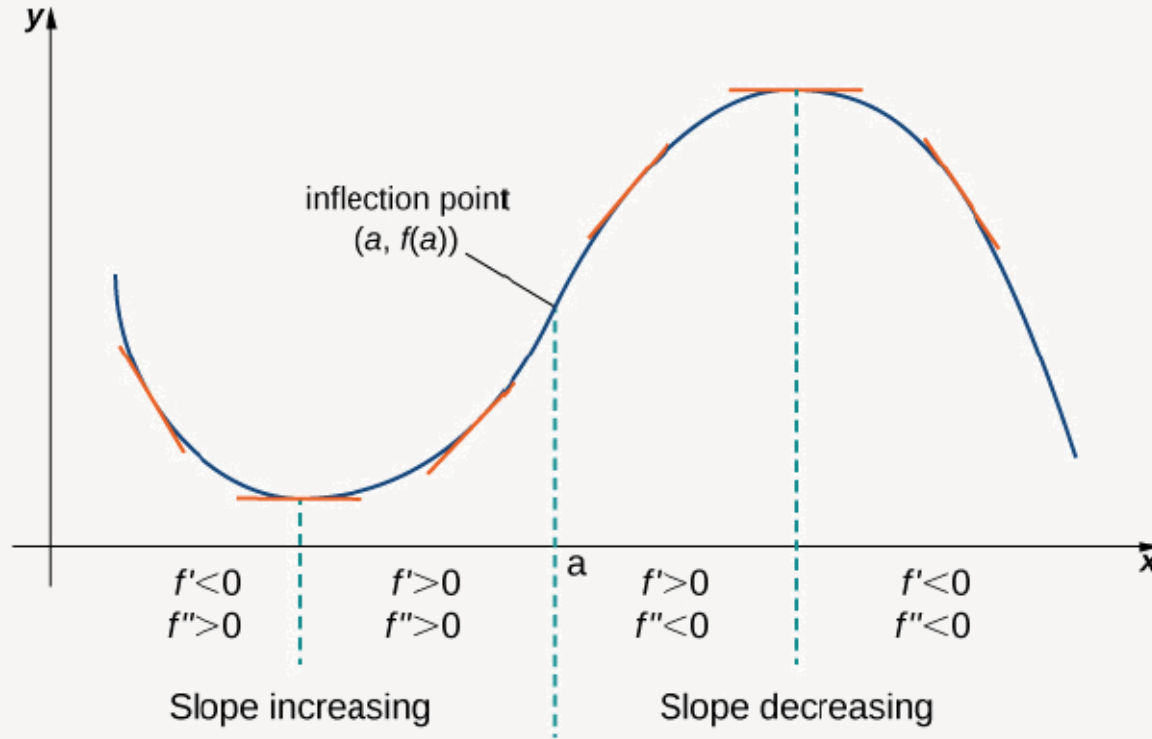
Assume f is a function that is at least twice differentiable, meaning f and f' are both differentiable.

Points where $f'(x) = 0$ are called stable points of f . Note that a function may have no stable points, a finite number of stable points, or an infinite number of them!

At a stable point x^* for f :

- If $f''(x^*) > 0$, the point is a local minimum.
- If $f''(x^*) < 0$, the point is a local maximum.
- If $f''(x^*) = 0$, we cannot determine the nature of the point based solely on the second derivative and must analyze higher-order derivatives.

Interpretation of First and Second Derivatives



Taylor series: Estimating a Function with a Polynomial

Assume that f is a well-behaved function, meaning it is infinitely differentiable (this is a very strong condition but can sometimes be relaxed). Also, assume that $x_0 \in \mathbb{R}$ is a fixed and desired point on the real number line.



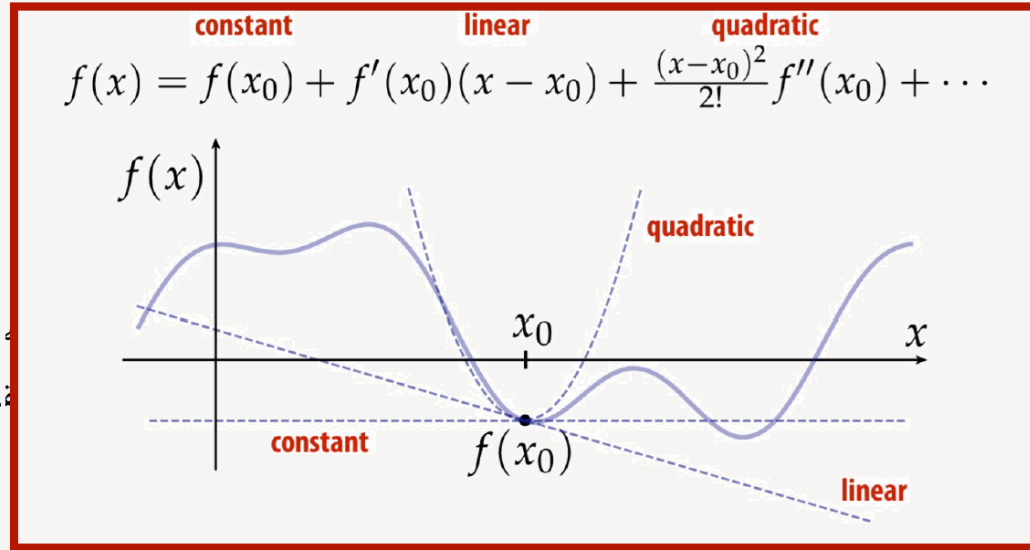
Under these conditions, and for some x (sometimes for all $x \in \mathbb{R}$), we have:

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k$$

The Taylor series of $f(x)$, even for points far away from x , provides an approximation of $f(x)$ based on the local information at the point x_0 .



Estimating a Function with a Polynomial



$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k$$

Taylor series Example

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

$$\sin(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$

$$x^3 - 3x + 1 = x^3 - 3x + 1$$

Taylor series Example

We consider the polynomial

$$f(x) = x^4 \quad (5.9)$$

and seek the Taylor polynomial T_6 , evaluated at $x_0 = 1$. We start by computing the coefficients $f^{(k)}(1)$ for $k = 0, \dots, 6$:

$$f(1) = 1 \quad (5.10)$$

$$f'(1) = 4 \quad (5.11)$$

$$f''(1) = 12 \quad (5.12)$$

$$f^{(3)}(1) = 24 \quad (5.13)$$

$$f^{(4)}(1) = 24 \quad (5.14)$$

$$f^{(5)}(1) = 0 \quad (5.15)$$

$$f^{(6)}(1) = 0 \quad (5.16)$$

Therefore, the desired Taylor polynomial is

$$T_6(x) = \sum_{k=0}^6 \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k \quad (5.17a)$$

$$= 1 + 4(x - 1) + 6(x - 1)^2 + 4(x - 1)^3 + (x - 1)^4 + 0. \quad (5.17b)$$

Multiplying out and re-arranging yields

$$T_6(x) = (1 - 4 + 6 - 4 + 1) + x(4 - 12 + 12 - 4) + x^2(6 - 12 + 6) + x^3(4 - 4) + x^4 \quad (5.18a)$$

$$= x^4 = f(x), \quad (5.18b)$$

i.e., we obtain an exact representation of the original function.

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Scalar Field Derivation



Scalar with respect to scalar

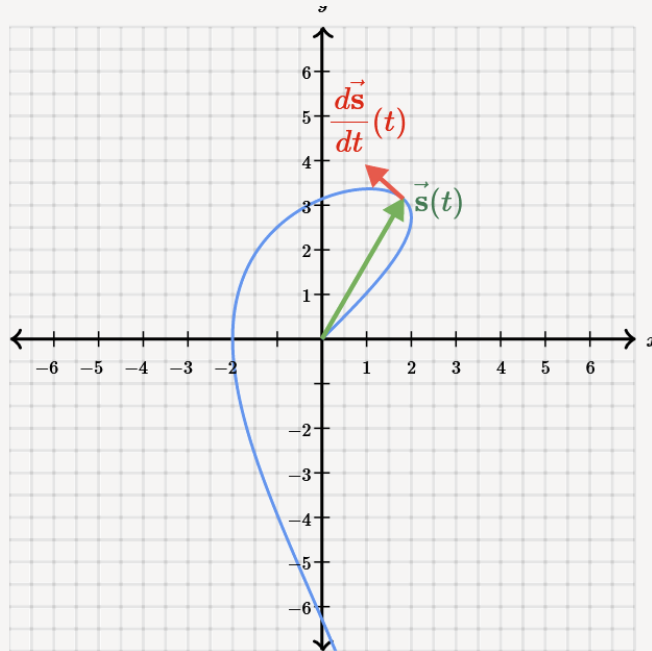


Vector-Valued Function

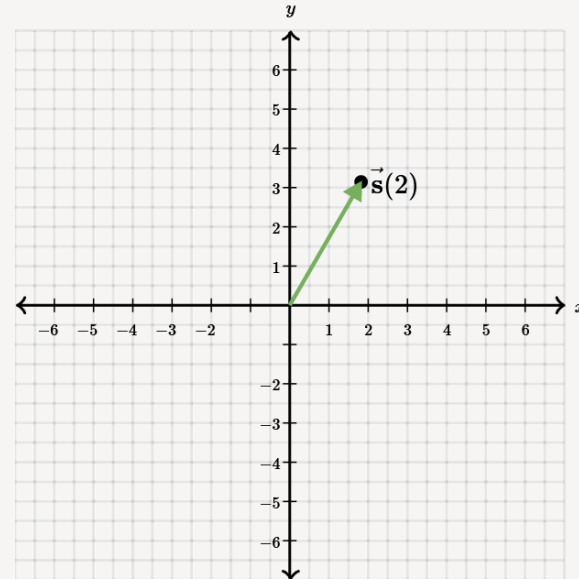


Directional Derivative

$$\vec{s}(t) = \begin{bmatrix} 2 \sin(t) \\ 2 \cos(t/3)t \end{bmatrix}$$

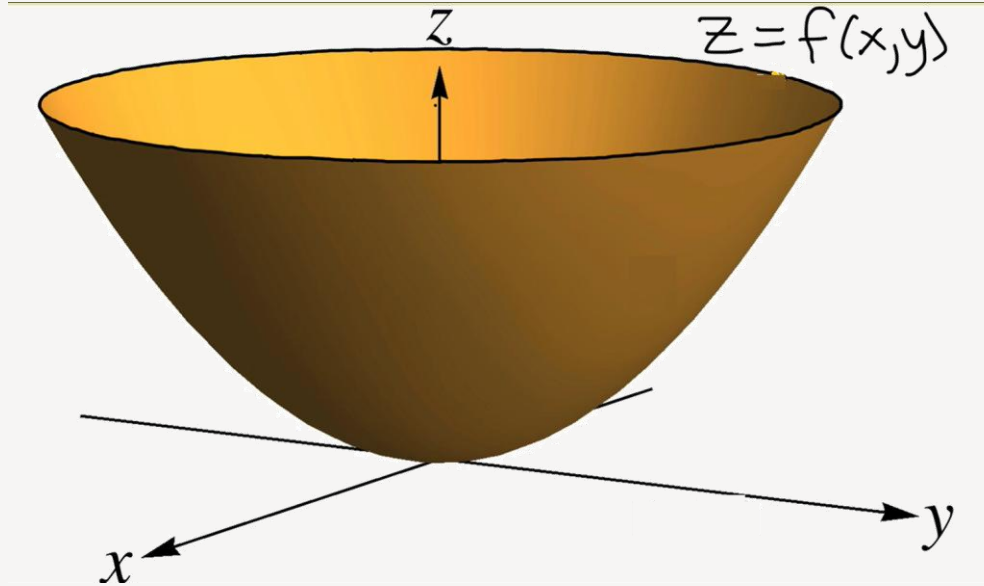


$$\vec{s}(2) = \begin{bmatrix} 2 \sin(2) \\ 2 \cos(2/3) \cdot 2 \end{bmatrix} \approx \begin{bmatrix} 1.819 \\ 3.144 \end{bmatrix}$$



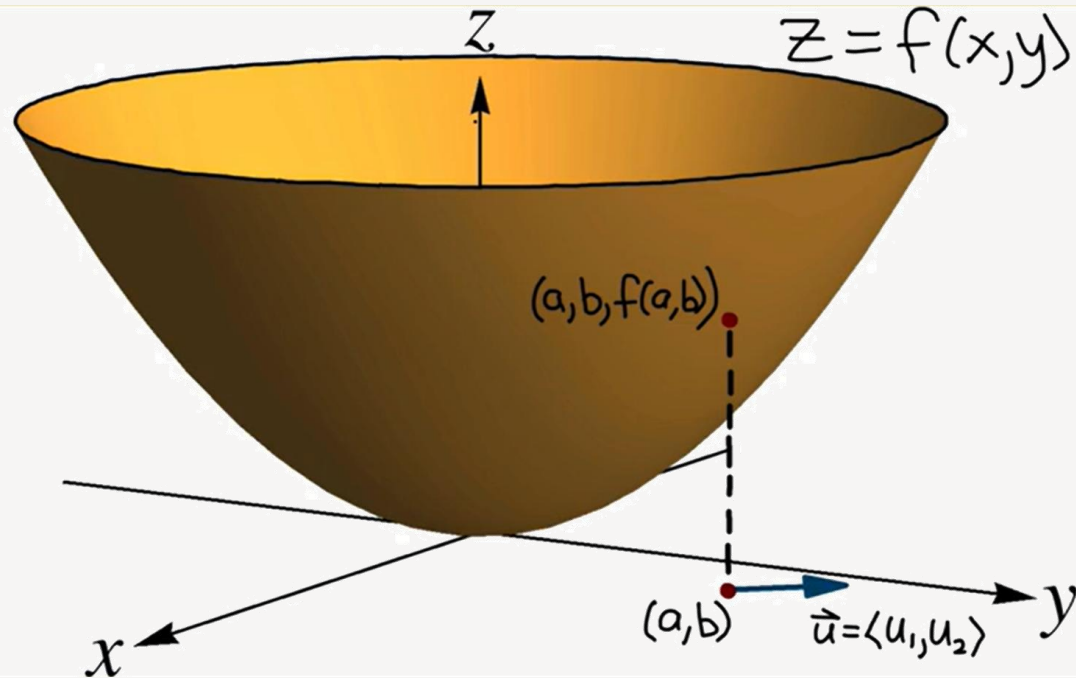
Directional Derivative

□ Example $D_{\vec{u}}f(a,b) = \vec{\nabla}f(a,b) \cdot \vec{u}.$



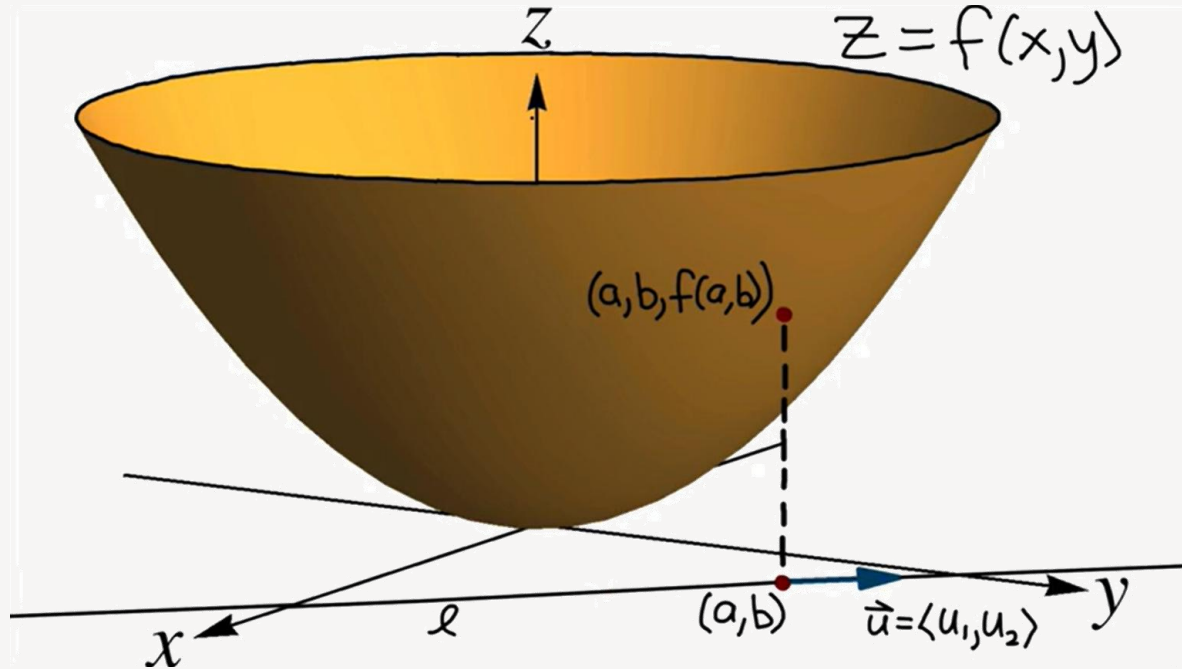
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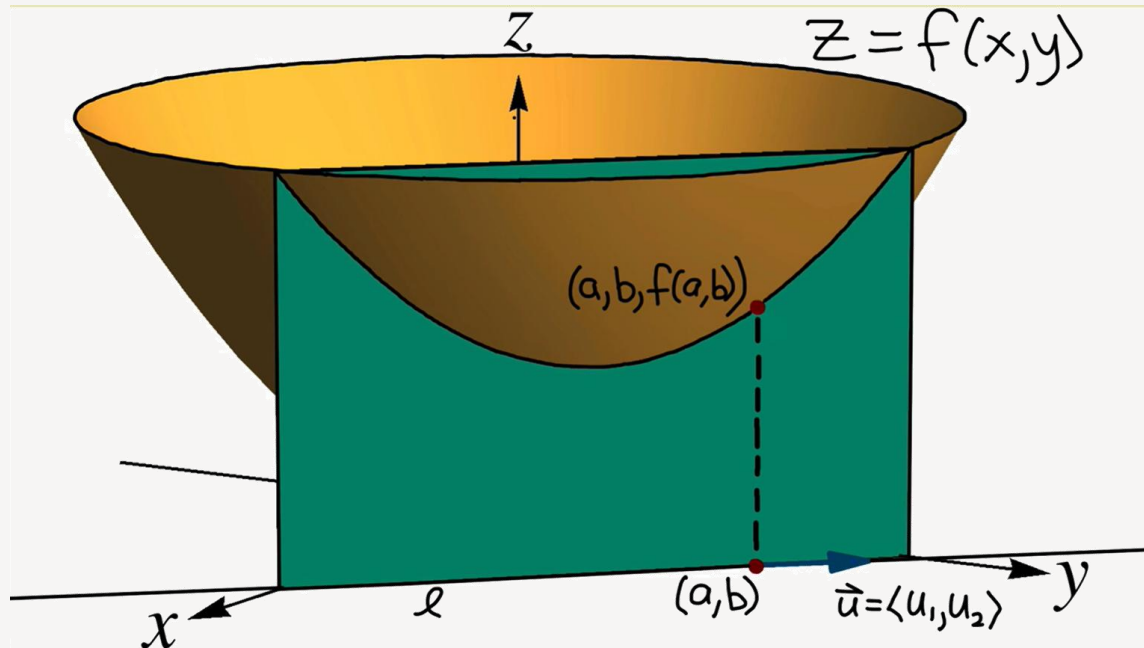
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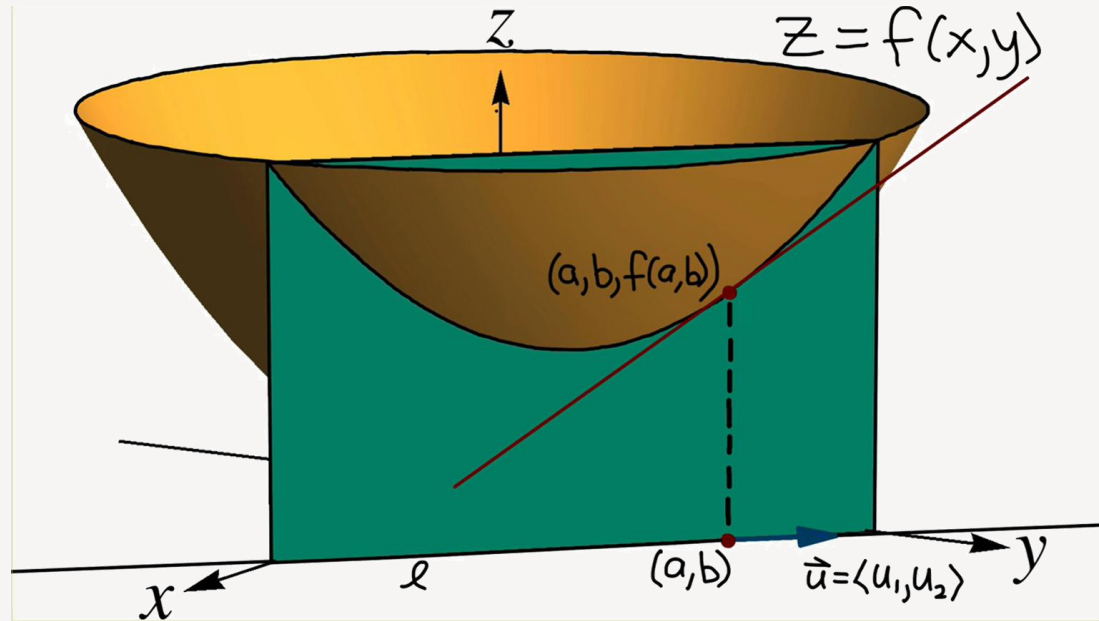
Directional Derivative

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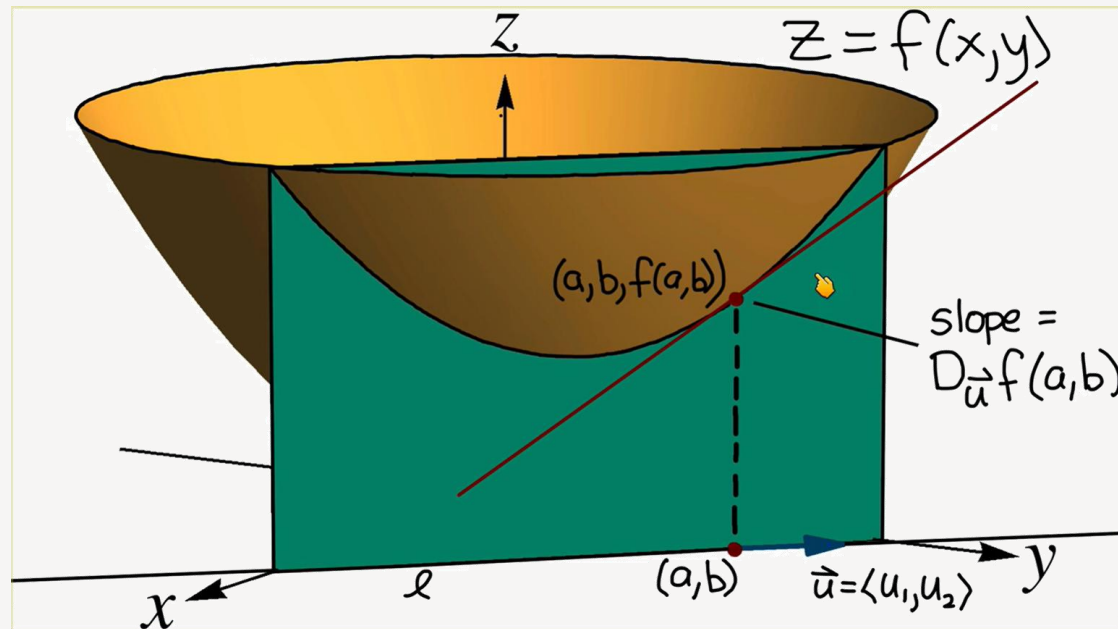
Directional Derivative

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Directional Derivative

□ Example $D_{\vec{u}}f(a,b) = \vec{\nabla}f(a,b) \cdot \vec{u}.$



Directional Derivative

$$f: \mathbb{R}^n \rightarrow \mathbb{R} \quad v = \begin{bmatrix} a \\ b \end{bmatrix}$$

$$D_v f = v \cdot \nabla f$$

$$\nabla_{\vec{v}} f(\mathbf{x}) = \lim_{h \rightarrow 0} \frac{f(\mathbf{x} + h\vec{v}) - f(\mathbf{x})}{h \|\vec{v}\|}$$

Scalar with respect to vector

Definition 5.5 (Partial Derivative). For a function $f : \mathbb{R}^n \rightarrow \mathbb{R}$, $\mathbf{x} \mapsto f(\mathbf{x})$, $\mathbf{x} \in \mathbb{R}^n$ of n variables $x_1, \dots, x_n$ we define the *partial derivatives* as

$$\begin{aligned} \frac{\partial f}{\partial x_1} &= \lim_{h \rightarrow 0} \frac{f(x_1 + h, x_2, \dots, x_n) - f(\mathbf{x})}{h} \\ &\vdots \\ \frac{\partial f}{\partial x_n} &= \lim_{h \rightarrow 0} \frac{f(x_1, \dots, x_{n-1}, x_n + h) - f(\mathbf{x})}{h} \end{aligned} \quad (5.39)$$

and collect them in the row vector

$$\nabla_{\mathbf{x}} f = \text{grad} f = \frac{df}{d\mathbf{x}} = \left[\frac{\partial f(\mathbf{x})}{\partial x_1} \quad \frac{\partial f(\mathbf{x})}{\partial x_2} \quad \dots \quad \frac{\partial f(\mathbf{x})}{\partial x_n} \right] \in \mathbb{R}^{1 \times n}, \quad (5.40)$$

The row vector in (5.40) is called the *gradient* of f or the *Jacobian*

Note!

Example

- $\frac{\partial(x^T a)}{\partial x} = a^T$

Remark (Gradient as a Row Vector). It is not uncommon in the literature to define the gradient vector as a column vector, following the convention that vectors are generally column vectors. The reason why we define the gradient vector as a row vector is twofold: First, we can consistently generalize the gradient to vector-valued functions $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ (then the gradient becomes a matrix). Second, we can immediately apply the multi-variate chain rule without paying attention to the dimension of the gradient.

Rules

Product rule:
$$\frac{\partial}{\partial x}(f(x)g(x)) = \frac{\partial f}{\partial x}g(x) + f(x)\frac{\partial g}{\partial x}$$

Sum rule:
$$\frac{\partial}{\partial x}(f(x) + g(x)) = \frac{\partial f}{\partial x} + \frac{\partial g}{\partial x}$$

Chain rule:
$$\frac{\partial}{\partial x}(g \circ f)(x) = \frac{\partial}{\partial x}(g(f(x))) = \frac{\partial g}{\partial f} \frac{\partial f}{\partial x}$$

Chain Rule

$$\frac{df}{dt} = \begin{bmatrix} \frac{\partial f}{\partial x_1} & \frac{\partial f}{\partial x_2} \end{bmatrix} \begin{bmatrix} \frac{\partial x_1(t)}{\partial t} \\ \frac{\partial x_2(t)}{\partial t} \end{bmatrix} = \frac{\partial f}{\partial x_1} \frac{\partial x_1}{\partial t} + \frac{\partial f}{\partial x_2} \frac{\partial x_2}{\partial t}$$

- **Example 1:**

Consider $f(x_1, x_2) = x_1^2 + 2x_2$, where $x_1 = \sin t$ and $x_2 = \cos t$, then

$$\begin{aligned} \frac{df}{dt} &= \frac{\partial f}{\partial x_1} \frac{\partial x_1}{\partial t} + \frac{\partial f}{\partial x_2} \frac{\partial x_2}{\partial t} \\ &= 2 \sin t \frac{\partial \sin t}{\partial t} + 2 \frac{\partial \cos t}{\partial t} \\ &= 2 \sin t \cos t - 2 \sin t = 2 \sin t (\cos t - 1) \end{aligned}$$

is the corresponding derivative of f with respect to t .



Chain Rule

$$\frac{df}{dt} = \begin{bmatrix} \frac{\partial f}{\partial x_1} & \frac{\partial f}{\partial x_2} \end{bmatrix} \begin{bmatrix} \frac{\partial x_1(t)}{\partial t} \\ \frac{\partial x_2(t)}{\partial t} \end{bmatrix} = \frac{\partial f}{\partial x_1} \frac{\partial x_1}{\partial t} + \frac{\partial f}{\partial x_2} \frac{\partial x_2}{\partial t}$$

- **Example 2:** If $f(x_1, x_2)$ is a function of x_1 and x_2 , where $x_1(s, t)$ and $x_2(s, t)$ are themselves functions of two variables s and t , the chain rule yields the partial derivatives

$$\begin{aligned} \frac{\partial f}{\partial s} &= \frac{\partial f}{\partial x_1} \frac{\partial x_1}{\partial s} + \frac{\partial f}{\partial x_2} \frac{\partial x_2}{\partial s}, \\ \frac{\partial f}{\partial t} &= \frac{\partial f}{\partial x_1} \frac{\partial x_1}{\partial t} + \frac{\partial f}{\partial x_2} \frac{\partial x_2}{\partial t}, \end{aligned}$$

and the gradient is obtained by the matrix multiplication

$$\begin{aligned} \frac{df}{d(s, t)} &= \frac{\partial f}{\partial \mathbf{x}} \frac{\partial \mathbf{x}}{\partial (s, t)} = \underbrace{\begin{bmatrix} \frac{\partial f}{\partial x_1} & \frac{\partial f}{\partial x_2} \end{bmatrix}}_{\frac{\partial f}{\partial \mathbf{x}}} \underbrace{\begin{bmatrix} \frac{\partial x_1}{\partial s} & \frac{\partial x_1}{\partial t} \\ \frac{\partial x_2}{\partial s} & \frac{\partial x_2}{\partial t} \end{bmatrix}}_{\frac{\partial \mathbf{x}}{\partial (s, t)}}. \\ &= \frac{\partial f}{\partial \mathbf{x}} \frac{\partial \mathbf{x}}{\partial (s, t)} \end{aligned}$$

Scalar with respect to matrix

The derivative of a scalar y by a matrix $X \in \mathbb{R}^{m \times n}$ is given by:


$$\frac{\partial y}{\partial X} = \begin{bmatrix} \frac{\partial y}{\partial X_{11}} & \frac{\partial y}{\partial X_{21}} & \cdots & \frac{\partial y}{\partial X_{m1}} \\ \frac{\partial y}{\partial X_{12}} & \frac{\partial y}{\partial X_{22}} & \cdots & \frac{\partial y}{\partial X_{m2}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial y}{\partial X_{1n}} & \frac{\partial y}{\partial X_{2n}} & \cdots & \frac{\partial y}{\partial X_{mn}} \end{bmatrix}$$


04

Vector Field Derivation



Vector with respect to vector

$$f : \mathbb{R}^n \rightarrow \mathbb{R}^m$$

For a function $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and a vector $\mathbf{x} = [x_1, \dots, x_n]^\top \in \mathbb{R}^n$, the corresponding vector of function values is given as

$$\mathbf{f}(\mathbf{x}) = \begin{bmatrix} f_1(\mathbf{x}) \\ \vdots \\ f_m(\mathbf{x}) \end{bmatrix} \in \mathbb{R}^m.$$

The differentiation rules for every f_i are exactly the ones we discussed in section 03

Why this happen??

$$= \begin{bmatrix} \frac{\partial f_1(\mathbf{x})}{\partial x_1} & \dots & \frac{\partial f_1(\mathbf{x})}{\partial x_n} \\ \vdots & & \vdots \\ \frac{\partial f_m(\mathbf{x})}{\partial x_1} & \dots & \frac{\partial f_m(\mathbf{x})}{\partial x_n} \end{bmatrix}$$

Vector with respect to vector

$$\frac{\partial f}{\partial x_i} = \begin{bmatrix} \frac{\partial f_1}{\partial x_i} \\ \vdots \\ \frac{\partial f_m}{\partial x_i} \end{bmatrix} = \begin{bmatrix} \lim_{h \rightarrow 0} \frac{f_1(x_1, \dots, x_{i-1}, x_i+h, x_{i+1}, \dots, x_n) - f_1(\mathbf{x})}{h} \\ \vdots \\ \lim_{h \rightarrow 0} \frac{f_m(x_1, \dots, x_{i-1}, x_i+h, x_{i+1}, \dots, x_n) - f_m(\mathbf{x})}{h} \end{bmatrix} \in \mathbb{R}^m$$

$$\begin{aligned} \frac{df(\mathbf{x})}{d\mathbf{x}} &= \begin{bmatrix} \boxed{\frac{\partial f(\mathbf{x})}{\partial x_1}} \dots \boxed{\frac{\partial f(\mathbf{x})}{\partial x_n}} \end{bmatrix} \\ &= \begin{bmatrix} \boxed{\frac{\partial f_1(\mathbf{x})}{\partial x_1}} \dots \boxed{\frac{\partial f_1(\mathbf{x})}{\partial x_n}} \\ \vdots \\ \boxed{\frac{\partial f_m(\mathbf{x})}{\partial x_1}} \dots \boxed{\frac{\partial f_m(\mathbf{x})}{\partial x_n}} \end{bmatrix} \in \mathbb{R}^{m \times n}. \end{aligned}$$

Jacobian Matrix

Vector with respect to scalar

$$f: \mathbb{R}^n \rightarrow \mathbb{R}^m$$

For a function $f: \mathbb{R}^n \rightarrow \mathbb{R}^m$ and a vector $\mathbf{x} = [x_1, \dots, x_n]^\top \in \mathbb{R}^n$, the corresponding vector of function values is given as

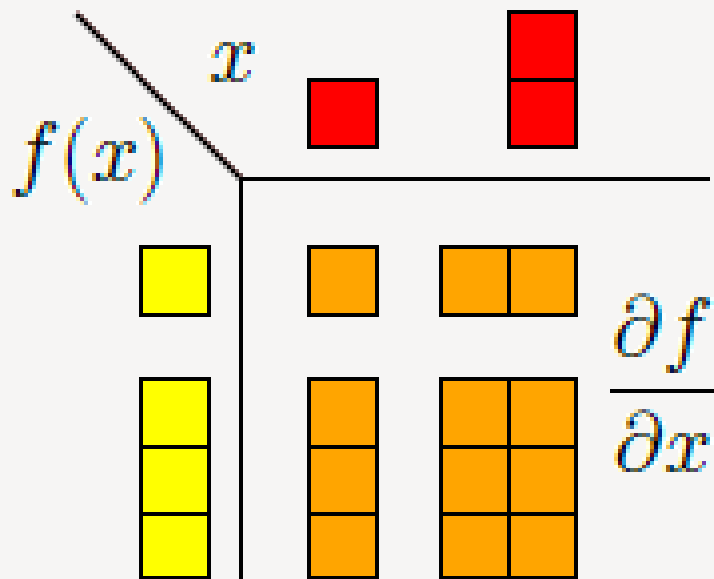
$$\mathbf{f}(\mathbf{x}) = \begin{bmatrix} f_1(\mathbf{x}) \\ \vdots \\ f_m(\mathbf{x}) \end{bmatrix} \in \mathbb{R}^m.$$

The differentiation rules for every f_i are exactly the ones we discussed in section 03

- If $x \in \mathbb{R}$ is a scalar, then it is a column vector

$$\begin{bmatrix} \frac{\partial f_1(\mathbf{x})}{\partial x} \\ \vdots \\ \frac{\partial f_m(\mathbf{x})}{\partial x} \end{bmatrix}$$

Dimensionality of (partial) derivatives



If $f : \mathbb{R} \rightarrow \mathbb{R}$ the gradient is simply a scalar (top-left entry).
 For $f : \mathbb{R}^D \rightarrow \mathbb{R}$ the gradient is a $1 \times D$ row vector (top-right entry). For $f : \mathbb{R} \rightarrow \mathbb{R}^E$, the gradient is an $E \times 1$ column vector, and for $f : \mathbb{R}^D \rightarrow \mathbb{R}^E$ the gradient is an $E \times D$ matrix.

Hessian Matrix

Suppose that $f(x) : \mathbb{R}^n \rightarrow \mathbb{R}$ is a function that takes a vector in $\mathbb{R}^n$ and returns a real number. Then the Hessian matrix with respect to x , written $\nabla_x^2 f(x)$ or simply as H is the $n \times n$ matrix of partial derivatives,


$$\nabla_x^2 f(x) = \begin{bmatrix} \frac{\partial^2 f(x)}{\partial x_1 \partial x_1} & \frac{\partial^2 f(x)}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f(x)}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f(x)}{\partial x_2 \partial x_1} & \frac{\partial^2 f(x)}{\partial x_2 \partial x_2} & \cdots & \frac{\partial^2 f(x)}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f(x)}{\partial x_n \partial x_1} & \frac{\partial^2 f(x)}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f(x)}{\partial x_n \partial x_n} \end{bmatrix}$$

In other words, $\nabla_x^2 f(x) \in \mathbb{R}^{n \times n}$, with

$$(\nabla_x^2 f(x))_{ij} = \frac{\partial^2 f(x)}{\partial x_i \partial x_j}$$

Note that the Hessian is always symmetric, since

$$\frac{\partial^2 f(x)}{\partial x_i \partial x_j} = \frac{\partial^2 f(x)}{\partial x_j \partial x_i}$$


05

Matrix Field Derivation



Matrix with respect to scalar

The derivative of a matrix $Y \in \mathbb{R}^{m \times n}$ by a scalar x is given by:


$$\frac{\partial Y}{\partial x} = \begin{bmatrix} \frac{\partial Y_{11}}{\partial x} & \frac{\partial Y_{12}}{\partial x} & \cdots & \frac{\partial Y_{1n}}{\partial x} \\ \frac{\partial Y_{21}}{\partial x} & \frac{\partial Y_{22}}{\partial x} & \cdots & \frac{\partial Y_{2n}}{\partial x} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial Y_{m1}}{\partial x} & \frac{\partial Y_{m2}}{\partial x} & \cdots & \frac{\partial Y_{mn}}{\partial x} \end{bmatrix}$$


06

Beautiful Examples!  

Important note on product Rule

Product rule:
$$\frac{\partial}{\partial \mathbf{x}} (f(\mathbf{x})g(\mathbf{x})) = \frac{\partial f}{\partial \mathbf{x}} g(\mathbf{x}) + f(\mathbf{x}) \frac{\partial g}{\partial \mathbf{x}}$$

Note. Please pay attention to following example!

- $$\frac{\partial (\mathbf{x}^T \mathbf{y})}{\partial \mathbf{z}} = \mathbf{x}^T \frac{\partial (\mathbf{y})}{\partial \mathbf{z}} + \mathbf{y}^T \frac{\partial (\mathbf{x})}{\partial \mathbf{z}}$$

- if $\mathbf{x}$ and $\mathbf{y}$ be vectors which elements are function of vector $\mathbf{z}$

Let's practice


$$\frac{\partial(u(x)+v(x))}{\partial x} = \frac{\partial u(x)}{\partial x} + \frac{\partial v(x)}{\partial x}$$


$$\frac{\partial(Ax)}{\partial x} = A$$


$$\frac{\partial(x^T a)}{\partial x} = a^T$$


$$\frac{\partial(x^T Ax)}{\partial x} = x^T (A + A^T)$$


$$\frac{\partial(x^T Ax)}{\partial x} = 2x^T A \text{ if } A \text{ is symmetric}$$

Hint!

$$A\vec{x} = \begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} a_1x_1 + a_2x_2 \\ a_3x_1 + a_4x_2 \end{bmatrix}$$

$$\frac{dA\vec{x}}{dx} = \begin{bmatrix} \frac{\partial(a_1x_1 + a_2x_2)}{\partial x_1} & \frac{\partial(a_1x_1 + a_2x_2)}{\partial x_2} \\ \frac{\partial(a_3x_1 + a_4x_2)}{\partial x_1} & \frac{\partial(a_3x_1 + a_4x_2)}{\partial x_2} \end{bmatrix}$$

$$= \begin{bmatrix} a_1 & a_2 \\ a_3 & a_4 \end{bmatrix} = A$$

Let's practice


$$\frac{\partial(A(t))^{-1}}{\partial t} = -A(t)^{-1} \frac{\partial(A(t))}{\partial t} A(t)^{-1}$$


$$\frac{\partial \det(A)}{\partial A} = \det(A) A^{-1}$$


$$\frac{\partial \ln(\det(A))}{\partial A} = (A^{-1})^T$$


$$\frac{\partial \det(A(t))}{\partial t} = \det(A) \operatorname{trace}\left(A^{-1} \frac{\partial(A(t))}{\partial t}\right)$$


$$\frac{\partial \operatorname{trace}(BA^{-1})}{\partial A} = -A^{-1}BA^{-1}$$


$$\frac{\partial(y^T Ax)}{\partial A} = yx^T$$


$$\frac{\partial(x^T Ax)}{\partial A} = xx^T$$

Review

Given $A = [a_{ij}]$, the (i, j) -cofactor of A is the number C_{ij} given by

$$C_{ij} = (-1)^{i+j} \det(A_{ij})$$

Then

$$\det(A) = a_{11}C_{11} + a_{12}C_{12} + \cdots + a_{1n}C_{1n}$$

Which is a cofactor expansion across the first row of A .

$$A^{-1} = \frac{1}{|A|} \begin{bmatrix} C_{11} & C_{21} & \cdots & C_{n1} \\ C_{12} & C_{22} & \cdots & C_{n2} \\ \vdots & \vdots & \ddots & \vdots \\ C_{1n} & C_{2n} & \cdots & C_{nn} \end{bmatrix} = A^{-1} = \frac{1}{|A|} \text{adj } A$$
$$\text{adj } (A) = C^T$$

The matrix of cofactors is called the **adjugate** (or **classical adjoint**) of A , denoted by $\text{adj } A$.

07

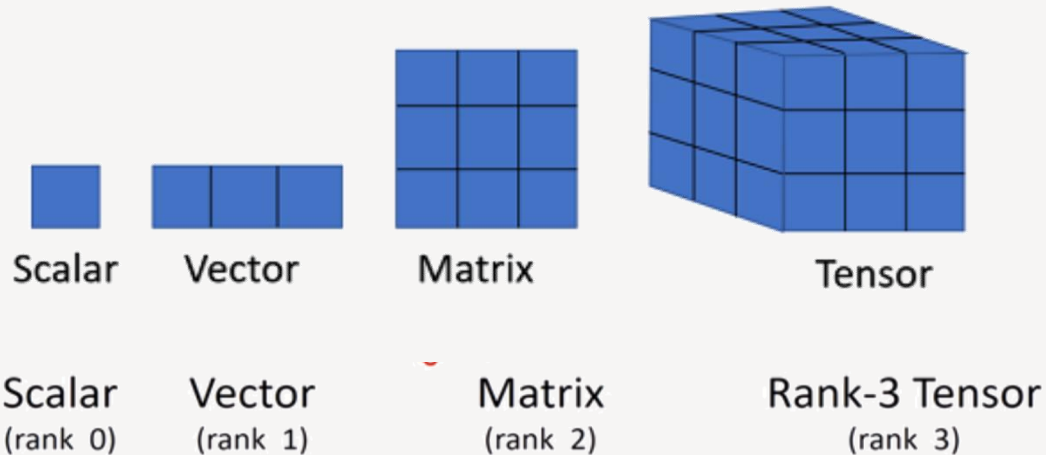
Tensors



Tensor

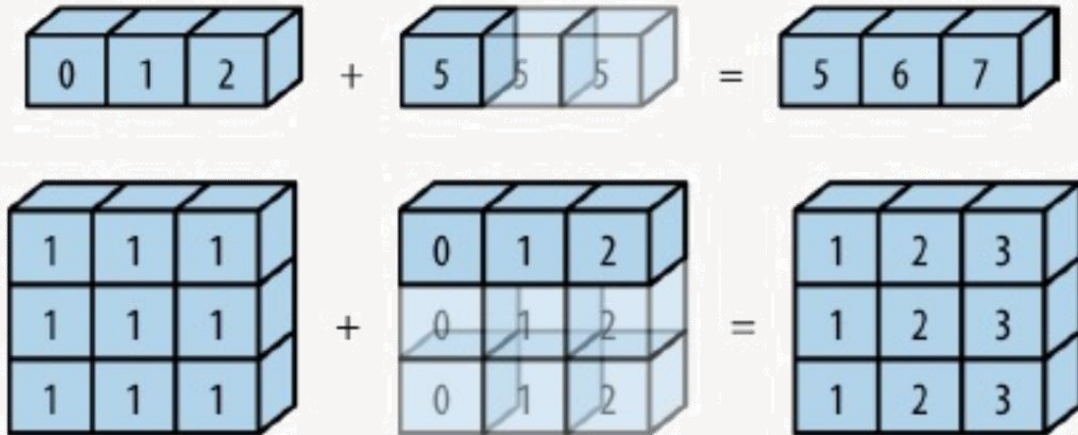
□ Multi-dimensional array of numbers

 `w = torch.empty(3)`
`x = torch.empty(2, 3)`
`y = torch.empty(2, 3, 4)`
`z = torch.empty(2, 3, 2, 4)`

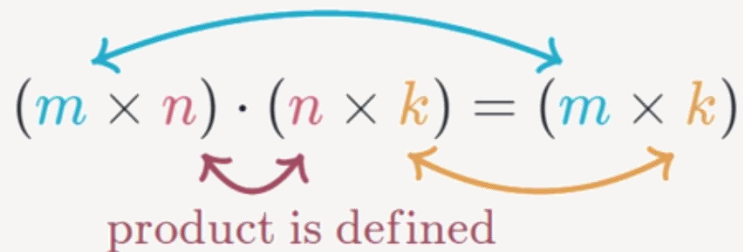


Tensors Addition

- Adding tensors with same size
- Adding scalar to tensor
- Adding tensors with different size: if **broadcastable**



Tensors Product



The diagram illustrates the compatibility of matrix dimensions for multiplication. It shows the equation $(m \times n) \cdot (n \times k) = (m \times k)$. A blue double-headed arrow connects the n in the first matrix's column dimension to the n in the second matrix's row dimension, indicating they must match. A red double-headed arrow connects the n in the first matrix's column dimension to the n in the second matrix's row dimension, with the text "product is defined" below it. An orange double-headed arrow connects the k in the second matrix's column dimension to the k in the resulting matrix's column dimension.

$$(m \times n) \cdot (n \times k) = (m \times k)$$

product is defined

Matrix with respect to vector

- Approach 1

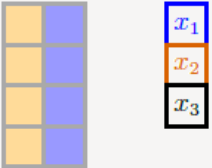
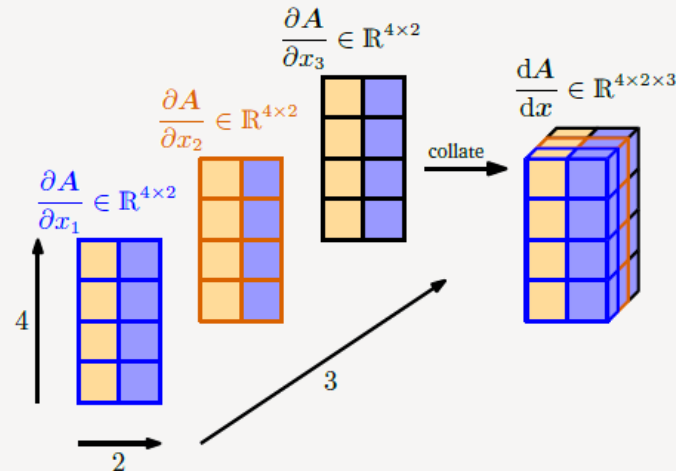
$$A \in \mathbb{R}^{4 \times 2} \quad x \in \mathbb{R}^3$$


Diagram illustrating the dimensions of the matrix A and vector x . Matrix A is 4×2 (4 rows, 2 columns). Vector x is 3×1 (3 elements: x_1, x_2, x_3).

Partial derivatives:

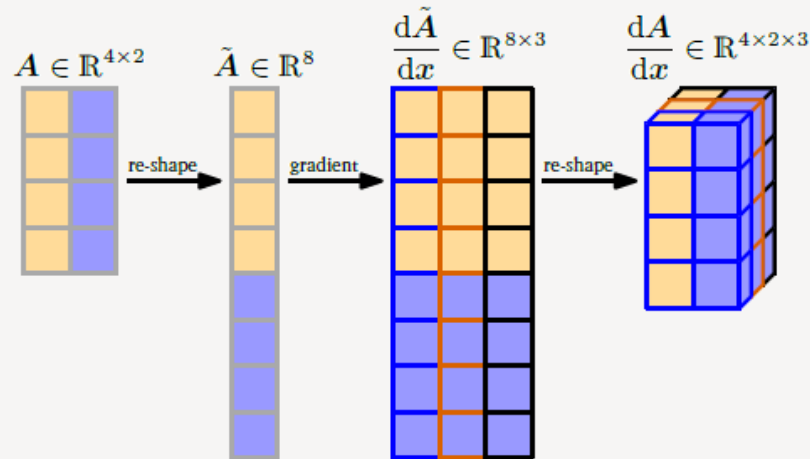


Matrix with respect to vector

- Approach 2

$$A \in \mathbb{R}^{4 \times 2}$$


$$x \in \mathbb{R}^3$$

References

- <https://explained.ai/matrix-calculus/>
- <https://paulklein.ca/newsite/teaching/matrix%20calculus.pdf>
- https://web.stanford.edu/~jduchi/projects/matrix_prop.pdf
- <https://www.math.uwaterloo.ca/~hwolkowi/matrixcookbook.pdf>
- https://www.kamperh.com/notes/kamper_matrixcalculus13.pdf

